

What we have, what would beat us, and the plan to close it

We are not starting from zero and we are not finished. This document is the honest position: an audit against the five judged metrics, the eight places a strong rival would out-argue us, and the five moves that would take this from a very good hackathon build to something that does not currently exist. **Every claim below is checked against a committed result file, not memory.**

01 Where we stand against the judged metrics

The weighting decides where effort belongs. Technical depth and problem fit are **55% between them**, so that is where the plan concentrates. Prototype quality is already our strongest asset and needs defending, not growing.

METRIC	WEIGHT	OUR POSITION	GRADE
Technical depth	30%		Strong
Identifiability derived, not asserted — three collinearities, two resolvable only by prior. Hand-written Kalman + RTS with exact likelihood, verified against an independent joint MVN to 1e−9. Physics validated non-circularly. Held back by: a compound specification error, a thermal layer that does not feed the headline number, and an evidence base of four circuits.			
Addresses the problem statement	25%		Strong, mis-aimed
The brief is <i>practice → clean curves → validate against race pace</i> . Experiment 03 is literally that. But our demo opens on a race, we use only FP2 and ignore FP1/FP3, and the practice→race pipeline has no dedicated product surface. We are answering the question in an appendix.			
Working prototype / feasibility	20%		Our best asset
Runs offline from one command, no Node and no API key. Nine views, 24 endpoints, live streaming replay, MCP server, 97 tests, verified from a clean clone. Most finalists will demo a notebook.			
Real-world applicability	15%		The soft flank
Adoption path and MCP integration are written and real. Missing the thing teams actually want: a forecast before they arrive at a circuit . No wet handling, no driver effects, no validation against real race outcomes.			
Presentation & communication	10%		Strong

36-page dossier, 24-slide deck, self-driving 4:55 demo video, README with screenshots, shareable web pitch. Needs re-cutting once the numbers below change — not rebuilding.

02 What we actually hold today

Estimator

Hierarchical linear-Gaussian state-space model in tyre age, fitted across the whole grid. Kalman filter + RTS smoother written in NumPy, six hyperparameters by L-BFGS-B on the exact likelihood.

9,802 lines Python · 38 modules

Evidence

Seven recorded experiments. Ground-truth recovery 0.0044 vs naive 0.0966 over 25 sessions. Practice→race 0.0518 vs 0.1166 over 5 events.

7 result files · every figure read, none typed

Product

Nine views, 3D racing line, live lap-by-lap replay at 0.22 ms/lap, hybrid retrieval over our own research corpus, MCP server with seven read-only tools.

6,531 lines TypeScript · 24 endpoints · 97 tests

And the number that should worry us

Our entire real-data claim rests on **10 comparisons across 5 events, all from 2024**. We hold 8 sessions covering 4 circuits. The 2024 season alone has **24 events**, and FastF1 exposes every one of them plus prior seasons. A rival who runs a full season has twenty times our evidence and will say so.

03 Where a strong rival would beat us

Ordered by how much damage each does under questioning.

01 Compound labels hide the physics TESTED – HYPOTHESIS NOT SUPPORTED

Pirelli nominates three of the C1–C5 range for each Grand Prix, so **"MEDIUM" at Monza is C4 while "MEDIUM" at Silverstone is C2 — different rubber**, and we pooled them as one compound. We predicted that was a specification error costing us accuracy. **Experiment 08 says otherwise**. Across 58 stint estimates from 23 races, pooling on the compound actually fitted transfers no better than pooling on the relative label — leave-one-event-out MAE 0.0499 against 0.0469, Wilcoxon $p = 0.16$. The likely mechanism is that Pirelli nominates to *equalise*: harder compounds for abrasive circuits, softer for gentle ones. The label already carries circuit severity, by design. The compound table is built and tested regardless, because the circuit-transfer model in M3 needs a physical compound to predict *for*.

02 **The evidence base is thin** **HIGH EXPOSURE**

Four circuits, one season, ten real comparisons. Every headline is defensible but under-powered. "Would this hold over a full season?" is the first question a statistician asks, and today the honest answer is "we don't know".

03 **We report the practice→race bias but never explain it** **OPEN**

Practice over-predicts race degradation by +0.047 s/lap in 9 of 10 comparisons. Reporting it honestly earns credit. **Solving it earns the win.** We hypothesised fuel load and never tested the cross-session version.

04 **The thermal and wear layer is unearned depth** **DECORATIVE**

A two-state thermal model and an energy-based wear formulation exist, are physically motivated, and feed nothing the headline estimate depends on. Either wire them into the estimator and show they improve a number, or present them as exploratory. Physics that changes no result is theatre.

05 **No pre-event capability** **THE MISSING PRODUCT**

A team's real question is "we arrive at a circuit on Thursday — what will the tyres do?" We can only answer once they have run. This is the gap between an analysis tool and a product.

06 **Lap-time intervals are undercovered** **CALIBRATION**

73% coverage against a nominal 95%, and pooled regression does better at 79%. We disclose it. We have not diagnosed it, and we have never drawn a PIT histogram or a reliability diagram.

07 **Rain is excluded, not handled** **SCOPE**

Wet and intermediate compounds are dropped at ingestion. Correct for now, indefensible for a product: a wet Friday is exactly when a team most needs help reading a dry race.

08 **No driver effect** **MODELLING**

Driving style materially changes degradation. We pool every car into one compound baseline with a stint-level deviation and never attribute any of it to the driver.

04 The five moves that decide this

Each states what it buys, which metric it moves, and — the part that matters — **the result that would prove it wrong**. A plan without falsifiable tests is a wish list.

M1 · Season-scale evidence

TECHNICAL DEPTH · PROBLEM FIT

Pull the full 2024 season — 24 events × FP1/FP2/FP3/Q/R — then 2023 and 2022. Re-run every experiment at scale. Ten practice→race comparisons become several hundred; four circuits become twenty-four. This is the foundation the other four moves stand on, and it is mostly download time rather than research risk.

Falsified if: error grows or interval coverage falls when the sample grows. That would mean our current numbers are a small-sample artefact — which we would then have to report.

M2 · Model the compound that was actually fitted

COMPLETE · NEGATIVE RESULT

Done, and the hypothesis did not survive. The per-event C1–C5 allocation table is built, cited row by row and covered by tests. But identity pooling does not beat label pooling on held-out events (0.0499 against 0.0469 s/lap, $p = 0.16$), and the mechanism is interesting on its own: Pirelli nominates to equalise, so the relative label is already a circuit-adjusted compound index. The table stays because M3 needs a physical compound to predict for — but the claim is now a published negative result, not a correction.

Falsified if: identity pooling does not beat label pooling on held-out events. **Outcome:** exactly that. Reported rather than retried until it agreed with us.

M3 · Predict a circuit we have never run

APPLICABILITY · TECHNICAL DEPTH

Regress compound degradation on circuit characteristics we already compute from GPS — per-corner frictional energy, lateral load spectrum, corner count, the share of the lap spent above a load threshold — and predict the degradation rate for a compound at a circuit absent from training. **This is the product.** It is the difference between explaining Friday and being useful on Tuesday.

Falsified if: leave-one-circuit-out prediction is no better than the global compound mean. That is the baseline to beat, and it is not a soft one.

M4 · Explain the practice → race bias, or characterise it properly

PROBLEM FIT · TECHNICAL DEPTH

Test the fuel-load hypothesis across sessions rather than within a stint, where we already proved it cannot work (per-lap energy varies only 2.3% inside a stint). If a fuel-weighted degradation clock removes the offset out-of-sample, the brief's central validation question goes from "honestly reported caveat" to "solved". If it does not, we will know why, on a season of data instead of five events.

Falsified if: the bias survives the correction on held-out events, or flips sign across the season — which would mean it is not a fuel effect at all.

M5 · Calibration rigour

TECHNICAL DEPTH

Replace coverage counting with proper probabilistic diagnostics: PIT histograms, reliability diagrams, sharpness against calibration. Diagnose the 73% lap-time undercoverage — most likely a heavy-tailed observation model fitted with Gaussian assumptions, which is testable. A model that reports intervals must be able to defend them.

Falsified if: the PIT histogram is uniform and the undercoverage turns out to be a scoring artefact rather than a model defect. That is a fine outcome — it just needs proving.

05 Sequence

Ordered by dependency, not preference. Nothing in Phase 2 can start before the data in Phase 0 exists. The durations assume focused work and will move; the *order* should not.

PHASE 0 Foundation

~1 day, mostly unattended

- **Season scrape.** All 24 events of 2024, every session type, cached as Parquet. Then 2023 and 2022. Runs unattended; start it first so it downloads while other work proceeds.
- **Compound allocation table.** Per-event C1–C5 nomination, committed as data with its source cited, plus a test that every event in the cache resolves.
- **Quality audit at scale.** Our exclusion rules were tuned on four circuits. Re-check the discard rate across twenty-four — a rule that drops 68% at Monza may drop 95% at Monaco.

PHASE 1 Re-establish every claim at scale

~2 days

- **Re-run experiments 01–07** on the full corpus. Publish the deltas honestly, including any that get worse.
- **Compound-identity re-specification (M2)** and the head-to-head against label pooling.
- **Full C-MAPSS test set.** Batch the fit per engine so all 100 engines score, removing the "indicative, not like-for-like" caveat.
- **Fix the practice emphasis.** The brief is about practice sessions; make FP1/FP2/FP3 first-class and lead the demo with them.

PHASE 2 The two new capabilities

~3 days · highest risk, highest return

- **Circuit transfer model (M3)** with leave-one-circuit-out validation, and a product surface that answers "what will C3 do here on Sunday" before anyone has run.
- **Practice→race bias resolution (M4)**, tested out-of-sample across the season.
- **Driver random effect.** Does attributing part of the stint deviation to the driver improve recovery? Cheap to add, and either answer is publishable.

PHASE 3 Rigour and honesty

~2 days

- **Calibration suite (M5)** — PIT, reliability, sharpness — surfaced in the product, not just the appendix.
- **Wet regime.** At minimum a detector that says "this model does not apply and here is why". Better: a wet degradation model, which nobody expects and everybody will notice.
- **Thermal layer decision.** Wire it into the estimator and prove it improves a number, or relabel it exploratory. No middle position.

PHASE 4 Product and story

~2 days

- **Practice→race workflow view** — the literal deliverable of the brief, as a first-class screen rather than a section inside "Does it work".
- **Cost of a wrong call**, quantified against real 2024 race outcomes rather than illustrative arithmetic.
- **Re-cut everything** — dossier, deck, video, README — against the new numbers. The self-driving demo regenerates from one command, so this is cheaper than it sounds.

06 If there is less time than the plan assumes

The order that maximises judged score per hour, if we have to stop early at any point.

#	DO THIS	METRIC WEIGHT	WHY IT IS FIRST
1	Season scrape	30 + 25	Unattended, and every other claim gets stronger for free. Start it before anything else.
2	Compound identity	30	Fixes a genuine specification error. Defensible as a contribution on its own.
3	Practice-first reframing	25	Cheapest large gain in the whole plan. We already do what the brief asks; we present it as an appendix.
4	Circuit transfer	15 + 30	The business case. Risky, so it needs a real time box — but nothing else differentiates us this much.
5	Calibration suite	30	Turns our weakest technical answer into a strength, and it is bounded work.
6	Re-cut the story	10	Last, because it depends on final numbers, and our presentation is already strong.

07 What could go wrong, and what we do about it

The season scrape is slow or rate-limited

FastF1 caches per session and telemetry is heavy.
Mitigation: start it first, run it unattended, and prioritise race + FP2 for all 24 events before adding FP1/FP3 and prior seasons. Partial coverage is still a large improvement on four circuits.

Results get worse at scale

Entirely possible — four circuits may have flattered us.
Mitigation: this is a feature, not a failure. Report the honest revised number and the reason. A model whose error is stable across 24 circuits at a slightly worse level is more credible than one that looks perfect on four.

Circuit transfer does not beat the compound mean

The highest-risk item. **Mitigation:** time-box it, and publish the negative result with the mechanism if it fails — as we did with the energy clock. A characterised failure is worth more than a silent retreat, and we already have a track record of doing exactly that.

Scope creep eats the presentation

The plan is deliberately larger than the time.
Mitigation: the priority ladder in section 06 is the contract. Everything below the line we reach is documented as future work with its falsifiable test attached, which reads as rigour rather than omission.

08

What "finished" looks like

Written now, so we are not deciding it under time pressure later.

METRIC	THE SENTENCE WE WANT TO BE ABLE TO SAY, AND DEFEND
Technical depth 30%	"We derived why this problem is unidentifiable, resolved what can be resolved, corrected a compound specification error nobody else models, and validated on a full season with proper probabilistic calibration — publishing the results that went against us."
Problem fit 25%	"From practice alone we produce a clean degradation curve per compound, and we validate it against race-day pace across a full season — which is precisely what the brief asked for, as the first screen of the product."
Prototype 20%	"One command, no network, no key. It runs live at 0.22 ms a lap and an agent can call it. Here is a clean clone doing it."
Applicability 15%	"We can tell a team what the tyres will do at a circuit before they arrive, and we can show the leave-one-circuit-out error on that claim."
Presentation 10%	"Every number on every slide is read from a committed file. Regenerate the experiments and the claims regenerate with them — including the ones that do not flatter us."

The one-sentence version, for the room

Everyone in this final will show a model that predicts tyre degradation. **We are the only ones who can show whether the reason our model gives is the right one** — and, by the end of this plan, the only ones who can predict a circuit we have never run.